



Constructing and interpreting volcano plots and activity maps to navigate homogeneous catalyst landscapes

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Volcano plots and activity maps are powerful tools for studying homogeneous catalysis. Once constructed, they can be used to estimate and predict the performance of a catalyst from one or more descriptor variables. The relevance and utility of these tools has been demonstrated in several areas of catalysis, with recent applications to homogeneous catalysts having been pioneered by our research group. Both volcano plots and activity maps are built from linear free energy scaling relationships that connect the value of a descriptor variable(s) with the relative energies of other catalytic cycle intermediates/transition states. These relationships must be both constructed and postprocessed appropriately to obtain the resulting plots/maps; this process requires careful execution to obtain meaningful results. In this protocol, we provide a step-by-step guide to building volcano plots and activity maps using curated reaction profile data. The reaction profile data are obtained using density functional theory computations to model the catalytic cycle. In addition, we provide volcanic, a Python code that automates the steps of the process following data acquisition. Unlike the computation of individual reaction energy profiles, our tools lead to a holistic view of homogeneous catalyst performance that can be broadly applied for both explanatory and screening purposes.

Introduction

Volcano plots and related activity maps are tools for predicting and rationalizing general trends surrounding catalytic behaviour^{1–3}. On the basis of Sabatier's principle⁴, which states that a catalyst should bind a substrate neither too strongly nor too weakly, volcano plots/activity maps estimate catalytic performance (e.g., overpotential, catalytic cycle thermodynamics/kinetics) from knowledge of a descriptor variable. This descriptor variable, a value that is used to characterize an individual catalyst, is typically the relative free energy of a single intermediate in the catalytic cycle and is selected on the basis of how well this quantity correlates with the relative free energy of other catalytic cycle intermediates and transition states. A volcano plot is obtained using one descriptor variable, while two or more are required to generate an activity map.

The foundations of both tools lie in the existence of linear free energy scaling relationships (LFESRs) that connect the value of the descriptor variable with the relative free energies of intermediates^{5,6} and transition states^{6,7} present in the catalytic cycle. Postprocessing these LFESRs leads to a volcano-shaped curve that reveals anticipated catalytic performance, with the best species appearing near the peak or plateau of the volcano. In the case of activity maps, postprocessing the LFESRs leads to a three-dimensional landscape, where the highest areas represent the location of the catalysts that have the highest anticipated activity. Individual points representing both known and prospective catalysts can then be added to the plot, which allows these tools to be used for both rationalizing observed catalytic performance and predicting the behaviour of untested catalysts.

Historically, volcano plots have predominantly been used to examine and predict the performance of catalysts for heterogeneous and electrocatalytic reactions. This ranges from the first volcano plots built using experimental data^{8,9} to more theoretical treatments using computed data^{10–17}. Despite the 60+ year history of volcano plots, only recently has the concept been transferred to the realm of homogeneous catalysis. In 2008, Swiegers¹⁸ proposed that Sabatier's principle should be valid for homogeneous systems, implying that the volcanoes could also be constructed for these types of

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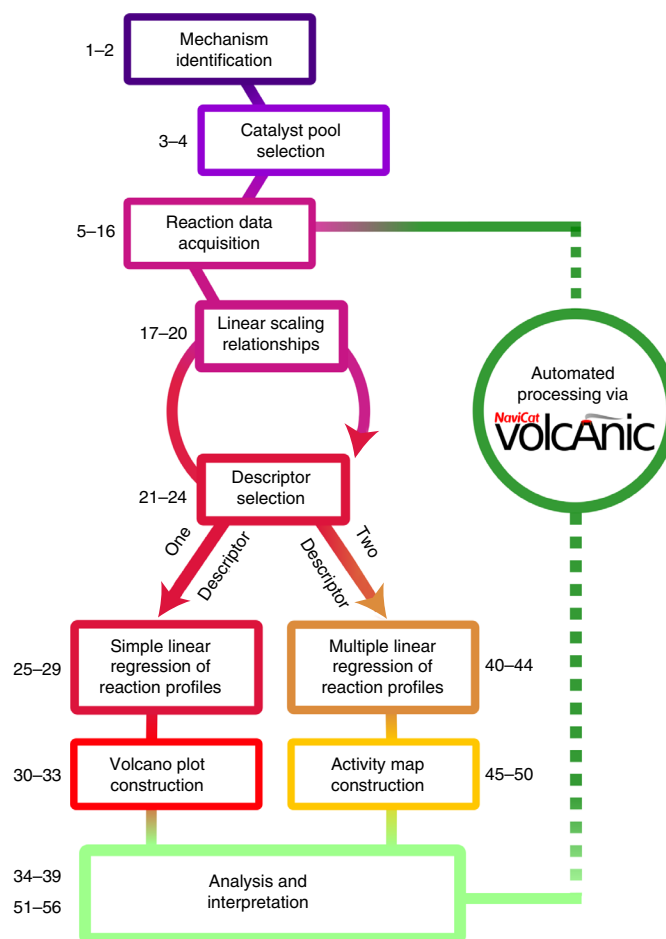


Fig. 1 | General workflow of the procedure. Schematic overview of the general workflow described in this protocol including the step numbers of the procedure.

catalytic transformation. However, no concrete proof-of-principle study was provided. In 2011, Morris¹⁹ related turnover frequency (TOF) with electronic and steric parameters of asymmetric transfer hydrogenation catalysts and constructed a type of volcano curve/activity map. The first proof-of-principle example of a Sabatier-type (i.e., energy-based) volcano plot for homogeneous catalysis came in 2015, when our research group constructed volcanoes for the Suzuki–Miyaura cross-coupling reaction⁶. Since that initial work, our research group has been at the forefront of analyzing and predicting the behaviour of homogeneous catalysts using volcano plots and activity maps³. Today, employing density functional theory (DFT) to model homogeneously catalyzed chemical reactions represents a straightforward route for obtaining the necessary reaction energy profiles to determine LFESRs and construct volcano plots and/or activity maps.

Overview of the procedure

In this protocol, we will describe how to construct plots and maps using computed data. In principle, it is possible to make use of experimental data, and we will comment on this in the ‘Limitations’ section.

Constructing a volcano plot/activity map requires sequentially executing eight key actions (Fig. 1) comprising 56 individual steps. To ease user burden, we have developed a Python code, volcAnic, which can be used to automate the protocol from Step 16 onward. The volcAnic package is freely available at <https://github.com/lcmd-epfl/volcAnic>. All data and instructions to reproduce each of the figures in this protocol are also included in the repository. In addition, here we supply the manual procedure in a step-by-step fashion. A fully worked-out example is available in Supplementary

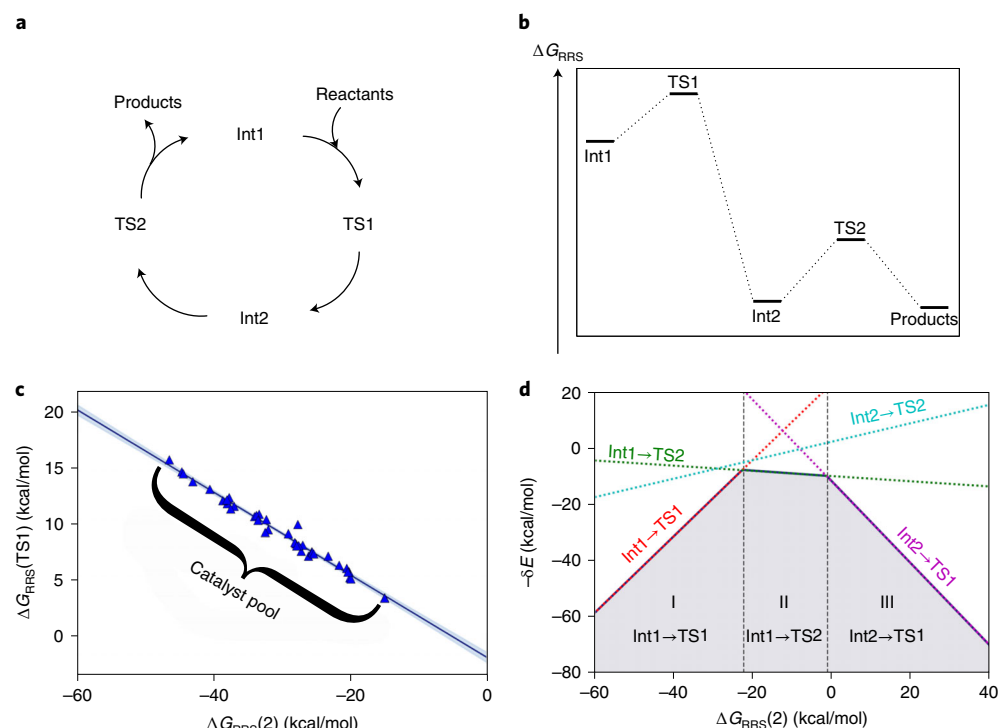


Fig. 2 | Schematic outline of the volcano plot construction process. **a**, Schematic representation of a catalytic cycle. **b**, Free energy profile of a representative catalyst for the catalytic cycle. **c**, Linear scaling free energy relationship relating the relative free energy of TS1 with the relative free energy of intermediate 2 for the entire pool of catalysts (blue triangles). **d**, Corresponding energy span SRP. The energy span volcano plot (shaded area in gray) arises from the lowest-lying line segments. The volcano plot is divided into regions I–III given by the identity of the limiting reaction steps for the corresponding descriptor variable range.

Methods. Note that Steps 1–16 assume experience in computational mechanistic modeling of catalytic reactions.

Mechanism identification

The catalytic cycle mechanism should be known and be robust so that catalysts possessing a variety of steric and electronic features each follow the same cycle. Computationally, this requires determining the relevant intermediates and transition states leading from reactants to products and establishing the relative Gibbs free energies of each species.

The necessary mechanistic information might be obtained from several sources, including insights made from experimental data or previous computational analysis. If no information is available, an exploration of plausible catalytic cycle pathways should be undertaken using a model catalyst. Strategies for identifying the appropriate catalytic cycle can be found elsewhere in the literature, and are beyond the scope of this protocol. After establishing a definitive catalytic cycle (Fig. 2a), one should proceed to the next step.

Catalyst pool selection

Select a medium-sized pool of sterically and electronically diverse catalysts (10–50 species) expected to follow the same catalytic cycle that will be used to establish the LFESRs.

Reaction data acquisition

Obtain the relative free energies with respect to a chosen reference state (ΔG_{RRS}) of all catalytic cycle intermediates (for thermodynamic volcanoes) or intermediates and transition states (for kinetic or energy span volcanoes) for each catalyst in the catalyst pool. By convention, the reference state is generally selected to be the separated catalyst and substrate(s); the combined energy of which is taken as ‘0’ for each catalyst. Computationally, this requires optimizing the geometries of reactants, intermediates (Int), transition states (TS) and products for each catalyst and establishing the relative

stabilities all the stationary points (Fig. 2b). Specific instructions for this process can be found elsewhere in the literature^{20–23}.

Linear scaling relationships and descriptor selection

Plotting the free energies of each intermediate/transition state pair against one another reveals underlying LFESRs that describe the energetic relationships between different species in the catalytic cycle (Fig. 2c). A key step is selecting an appropriate descriptor variable, which is generally the relative free energy of the catalytic cycle intermediate that provides the most accurate description of the relative energies of the other intermediates/transition states. Typically, the energies of all reaction steps are linearly regressed as a function of all possible descriptors, and the final decision on the most appropriate descriptor variable is established by the overall quality of the linear fits. Assuming that there are good LFESRs, the energetics of the complete catalytic cycle can be estimated directly from knowledge of this descriptor variable and the corresponding LFESRs.

Creating SRPs and volcano plots as a function of the descriptor variable

The LFESRs obtained from the previous step can predict the relative free energy of each catalytic cycle intermediate/transition state (Fig. 2a) based on the value of the descriptor variable (e.g., connected reaction steps or the difference between possible resting states and possible rate-limiting transition states). The LFESRs can subsequently be postprocessed to reveal the free energy change associated with moving between two stationary points in the catalytic cycle. Plotting these new linear relationships as a function of the descriptor variable leads to a simulated reaction profile (SRP, Fig. 2d). Obtaining the final volcano plot (gray area, Fig. 2d) involves taking the most energetically costly step for each value of the descriptor variable. A similar process using two, as opposed to one, descriptor variables leads to the creation of an activity map.

Analysis of the volcano plots/activity maps and interpreting results

Analyzing volcano plots/activity maps reveals general trends in catalytic behaviour. In a volcano plot, the value of the descriptor variable is plotted along the *x*-axis, while the *y*-axis reflects a measure of catalytic activity (the energy associated with the most difficult reaction step, the energy span, or the TOF). The volcano is constructed such that catalysts appearing high on the plot (near the peak or plateau region) have high relative activity, while species found lower on the plot (along the slopes) are predicted to be less active. For activity maps, two descriptor variables are generally used (plotted on the *x*- and *y*-axis), while activity is displayed on the *z*-axis, often in the form of a heat map.

Assessing the performance of individual catalysts involves adding points to the plot for each catalyst. This is accomplished using the value of the descriptor variables (*x*-coordinate for volcano plots, *x*- and *y*-coordinates for activity maps) and the chosen measure of catalytic activity (*y*-coordinate for volcano plots, *z*-coordinate for activity maps) as a set of Cartesian coordinates, which reveals the relative activities of each species. Comparing the activities of the different catalysts can be used to rationalize (experimentally) observed behaviour and to understand the chemical factors that lead to enhanced activity. A second use of volcano plots/activity maps is screening previously untested catalysts to identify highly active species and/or regions of catalyst space where highly active species are found. Here, volcanoes/activity maps offer a convenient tool for quickly estimating anticipated catalytic performance whereby, knowing only the value of the descriptor variable(s) for an untested catalyst, an estimation of activity can be obtained.

Applications of the method

Volcano plots and activity maps can be applied to nearly any catalytic reaction, provided the existence of acceptable-quality LFESRs and a robust catalytic cycle. In 2015, our research group constructed volcanoes for the Suzuki–Miyaura cross-coupling of vinyl bromide and vinylboronic acids to form butadiene⁶. Using a pool of 36 transition metal catalysts, a thermodynamic volcano (i.e., a volcano where only the energies associated with moving between catalytic cycle intermediates are considered) was constructed that reproduced the experimental trends surrounding the relative performance of different metals (i.e., Pd, Pt > Ni > Au, Cu, Ag). The robustness of the molecular volcano plot concept was subsequently confirmed by showing that catalysts with different oxidation, spin and ligation states all follow the same volcano curve²⁴.

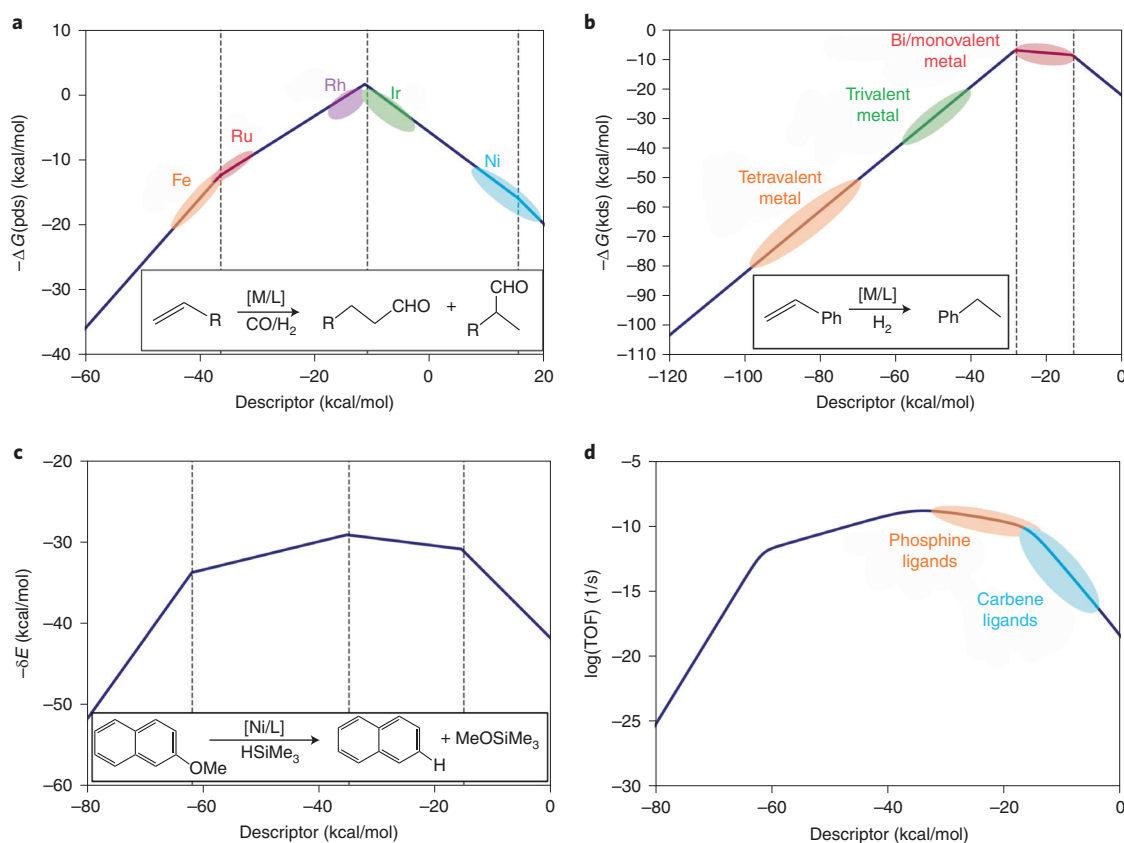


Fig. 3 | Examples of catalytic processes that have been studied using different variants of the volcano plots. a, Thermodynamic volcano for alkene hydroformylation. **b,** Kinetic volcano for the alkaline earth-metal catalyzed hydrogenation of alkenes. **c,** Energy span volcano for nickel-catalyzed aryl-ether cleavage. **d,** TOF volcano for nickel-catalyzed aryl-ether cleavage.

Since our proof-of-concept thermodynamic case study (for an example of a thermodynamic volcano plot, see Fig. 3a), we have made a series of refinements aimed at more accurately reproducing experimental data, first by including kinetic aspects of the catalytic cycle²⁵ followed by directly predicting the TOF of catalysts within the volcano framework (see also Steps 31–33 of the Procedure for an explanation of how the TOF can be calculated using this approach)^{26,27}.

This was complemented by extensions that addressed pertinent issues including aspects of selectivity²⁸ and other catalytic features of interest to experimentalists, such as substrate scope²⁹. Some of our more recent work has focused on increasing number of catalysts that can be screened by machine-learning the descriptor variable³⁰ such that hidden chemical trends can be uncovered using data-mining techniques³¹. Most recently, we have created more accurate ways of accounting for all thermodynamic aspects of the catalytic cycle, through the creation of ‘augmented’ volcanoes³².

Each of the aforementioned developments applies to ‘two-dimensional’ volcano plots, where a single descriptor variable is related to the catalytic property of interest. Note that, a second descriptor can be included to create ‘three-dimensional’ activity maps. Proceeding in this manner, for example, allowed us to create activity maps that simultaneously provide the thermodynamics of multiple cross-coupling variants (Suzuki–Miyaura, Kumada, Negishi, Stille, Hiyama, etc.)³³ as well as examine the influence of the electrophilic coupling component in Suzuki–Miyaura C–C coupling³⁴.

To complement the more conceptual advancements discussed above, we and others have employed the volcano plot/activity map framework in the study of interesting chemical problems^{27,31,33,35,36}. One notable example gauged the influence of oxidation state and metal size for the alkaline earth-metal catalyzed hydrogenation of alkenes. Here, the valence of the metal atom was found to be crucial, with low-valence species (mono and bivalent) making good catalysts, while higher-valence species (tri- and tetravalent) were found to be worse catalysts (Fig. 3b)³⁷. A second

example uses a machine-learning/volcano plot combination to examine the activity of >143,000 catalysts for the reductive cleavage of C(sp²)-O bonds in aryl ethers by nickel catalysts³⁸. Examination of both phosphine and carbene ligands revealed that neither ligand class is ideally tuned to catalyze this reaction (Fig. 3c,d). However, incorporation of certain structural factors led to the identification of Ni-phosphine catalysts predicted to exhibit high activity.

For the interested reader, a more comprehensive description of the different volcano plot variants as well as descriptions of relevant examples is provided in a recent account³.

Limitations

Volcano plots and activity maps suffer from two main limitations.

The approach requires good LFESRs

The first is constructing a map/plot when the LFESRs are of poor quality. While volcano plots are generally robust against outliers owing to the statistical nature of the LFESRs, if there are no clear linear relationships, the resulting volcanoes are susceptible to inaccurate predictions. This problem principally arises from either a complete lack of scaling relationships among the data or when the chosen catalysts pool used to establish the LFESRs is not sufficiently diverse (*i.e.*, the data is clustered). A lack of clear scaling relationships is difficult to remedy. However, building an activity map using two descriptor variables may alleviate the problem by having one variable describe one portion of the catalytic cycle and a second variable describe the remaining portion of the catalytic cycle^{12–14}. Clustering of the catalyst points can often be overcome through the addition of more species and recalculating the LFESRs. This procedure can be done iteratively until sufficiently robust LFESRs are obtained.

The catalytic mechanism needs to be correct and the same for all the tested catalysts

Problems regarding deviations from the expected catalytic mechanism cannot be explicitly handled within the volcano plot/activity map framework. Ideally, any catalysts selected as potential experimental targets from volcano plots would be subject to a more rigorous theoretical analysis, where potential deactivation and alternative pathways would be fully characterized. In this sense, volcano plots and activity maps provide a type of ‘best case’ scenario, where each of the catalysts examined is expected to follow a prescribed reaction pathway. The identification of off-cycle products (*e.g.*, as detected by experiment) could be used to create additional volcano plots/activity maps in which the corresponding reactions are included using either computational or experimental data. If the relative stability of an off-cycle product can be established, the corresponding line or surface can be included in the SRP (Steps 25–29). If used together with the original plot, the new plots could identify strategies for steering reactions toward the desired product.

Comparison with other methods

Obtaining the energetics of homogeneous catalytic processes involves computing the intermediates and transition states of an anticipated catalytic cycle using a standard computational technique, such as DFT. In turn, this information can be used to assess global catalytic activity and selectivity of a single catalyst/substrate combination. Repeating this process for different catalyst/substrate combinations would provide information about the relative performance of the different catalysts. These types of tasks, however, are limited to analyzing only a handful of catalysts. While the information obtained about each system is detailed, the ability to abstract design principles leading to more active/selective catalysts is limited.

In contrast, using volcano plots/activity maps offers a wealth of advantages over the aforementioned standard computational method. Notably, the volcano plot provides the area of anticipated maximum catalytic activity. This fact, combined with the array of steric and electronic properties of catalysts in the pool used to construct the LFESRs, allows the user to judiciously choose the appropriate steric/electronic ligand elements and metal center to maximize activity. Rational design principles of this type are not obtained using the standard ‘free energy profile’ technique that is common throughout the literature.

Experimental design

The main choice for the user is selecting an appropriate theoretical level to compute the reaction free energy profiles of the catalytic cycle. In principle, any desired level can be used to construct volcano

plots/activity maps, so long as the LFESRs are sufficiently robust. The specific theoretical level employed should be adjusted to best reflect the chemistry of the problem of interest. Some of our previous work has used the following computational procedure, which represents a good starting point for a first-time user:

- Optimizations at the M06/def2-SVP level (using Gaussian)
- Single point energies at the PBE0-dDsC/TZ2P level (using ADF)^{37–41}
- Free energy corrections obtained from the M06/def2-SVP computations (Gaussian)
- Solvation corrections using COSMO-RS (ADF)^{24,27,36,42}

If a change in molecularity occurs during the catalytic cycle, it is important to add a correction for the change of entropy in order to correctly estimate the free energies of association and dissociation processes, respectively. Previously, we have accomplished this using the technique of Martin, Hay and Pratt⁴³. For a comprehensive discussion of methods for treating the entropy of chemical processes in solutions, see the supporting information of Gallarati et al⁴⁴.

Materials

- Computer with a standard operating system. Additional computing facilities may be required for Steps 1–15. For example: more computational infrastructure (i.e., a computing cluster) will facilitate the study of multiple catalytic systems in a reasonable timeframe. This is especially true when ‘large’ catalysts are being studied
- Quantum chemistry software with basic functionalities (e.g., Gaussian⁴⁵, ADF⁴⁶, ORCA⁴⁷ and PSI4⁴⁸)
- Molecular geometry manipulation software (e.g., GaussView⁴⁹ and Avogadro⁵⁰)
- A basic plotting tool (e.g., Python with Matplotlib⁵¹, GNUplot, MATLAB and MS Excel) to follow the manual procedure
- Python 3 package (<https://www.python.org/download/releases/3.0/>) to use volcanic and its dependencies to follow the automated procedure
- volcanic (<https://github.com/lcmd-epfl/volcanic>) is required to follow the automated procedure described in Box 1. Table 1 in Supplementary Method shows an example of a sample input file for volcanic

Box 1 | Automated processing via volcanic ● Timing Minutes.

Specific instructions for the examples shown in all figures of this protocol are provided in Supplementary Methods. The following example generates the volcano plots in Fig. 3c,d.

Procedure

- 1 Download the volcanic package from <https://github.com/lcmd-epfl/volcanic>.
- 2 The free energy reaction profiles of the aryl-ether cleavage reaction for 21 catalysts are tabulated and stored in a file called aryl_ether_cleavage.xlsx available in the examples directory. Execute:

```
python volcanic.py -i examples/aryl_ether_cleavage.xlsx
```

- 3 Answer yes/no (y/n) to the questions in the prompt to generate a volcano plot/activity map from your data. For instance, answer

Generate energy span volcano plot ? (y/n):

with ‘y’ to generate Fig. 3c, and answer

Generate TOF volcano plot ? (y/n):

with ‘y’ to generate Fig. 3d. Several plots/maps can be generated in one run. After choosing the type of volcano plot/activity map to build, volcanic will propose a plausible descriptor variable:

ΔGRRS(3) has been identified as a suitable descriptor variable with $r^2 = 0.8211$. Continue using this variable ? (y/n):

Input ‘y’ to select the descriptor variable. The user is guided throughout the procedure and is only required to input y/n.

- 4 A comprehensive list of options can be obtained by running:

```
python volcanic.py -h
```

Command line arguments can be passed to bypass manual input. Examples and documentation are available in the repository.

Procedure

Mechanism identification ● Timing Days to weeks, depending on the number of steps in the catalytic cycle and on the existence of previous literature reports

- 1 Identify the catalytic cycle mechanism using experimental and/or computational data.
- 2 If computing the mechanism, use a representative model or experimentally known catalyst.
▲ CRITICAL STEP For thermodynamic volcanoes or activity maps, only the intermediate species (i.e., not transition states) are required.

Catalyst pool selection ● Timing Minutes to hours

- 3 Identify the metal center and the coordinated ligands.
- 4 Construct a pool of theoretically possible catalysts by varying the metal and the coordinated ligands. Ideally, the catalyst pool should consist of 10–50 species.
▲ CRITICAL STEP When changing the metal center, use elements from the same group or an isoelectronic equivalent. Changing metal groups will result in a change of charge of the overall system. For example, a neutral-charged Pd catalyst would carry a -1 charge with Rh and a $+1$ charge with Ag. The same considerations apply to replacing ligands having different charges. Changing the overall charge of the system may introduce unforeseen mechanistic changes and a corresponding breakdown of the volcano concept.
▲ CRITICAL STEP Ensure sufficient diversity in the catalyst pool to obtain robust scaling relationships. Ideally, the catalyst pool should explore a range of electronic effects (e.g., spanning from electron donating to electron withdrawing substituents within ligands), steric effects (e.g., introducing groups of different sizes in ligands) and hard–soft sites (e.g., going from softer to harder metal centers) to ensure that the descriptor variable will cover a wide range of relative free energy values.

Reaction data acquisition ● Timing One to several weeks, depending on catalyst size and the number of catalytic cycle intermediates and transition states. This is the most time-consuming step in creating volcano plots/activity maps. The timing assumes sufficient computational resources and a cost of 10^4 – 10^5 CPU h.

- 5 Use a molecular geometry manipulation tool to build the structures of each catalytic cycle intermediate/transition state for all the catalysts in the pool. In our work we typically use GaussView⁴⁹.
- 6 Run full geometry optimizations on each catalytic cycle intermediate. In our work we typically use Gaussian⁴⁵.
- 7 Run transition state optimizations on each catalytic cycle transition state. Note that running a constrained optimization (where the key bond lengths/angles of a previously optimized transition state are frozen) before a full transition state optimization may lead to more facile convergence.
- 8 After the intermediate/transition state structures are converged, run frequency calculations (at the same theoretical level used in the previous step) for all species. In many programs, the optimization and frequency steps can be combined together in one input file.
- 9 If desired, run single point computations on the optimized geometries.
- 10 Confirm that all the calculations have terminated normally. Obtain the total energy from the optimization or single point at the highest theoretical level.
- 11 Obtain the thermochemical corrections from the frequency computations.
- 12 If needed, obtain the solvent corrections for all the species in the relevant solvent.
- 13 If necessary, correct the translational and rotational entropies using the method of Martin, Hay and Pratt⁴³ (or another selected alternative). This step is particularly important if any steps of the catalytic cycle include a change of molecularity.
- 14 Determine the solvent-corrected free energy values by summing the electronic energy from the highest level of theory (single point or optimization), the thermal correction to the Gibbs free energy (from the frequency computation), and the solvation correction (if not already included in the electronic energy).
- 15 Calculate the relative stabilities, $\Delta G_{\text{RRS}}(\text{Int/TS } n)$, of each intermediate and transition state in the catalytic cycle. Free energies should be computed relative to the separated catalyst and substrate. The ΔG value for any intermediate or transition state n (Int/TS n) is determined by:

$$\Delta G_{\text{RRS}}(\text{Int/TS } n) = [G(\text{Int/TS } n) - (G(\text{Cat}) + G(\text{Substrate}))]$$

- 16 Tabulate the final free energy profiles for each catalyst in a spreadsheet. Arrange the relative free energy profiles of each species from the catalyst pool such that each row corresponds to a different catalyst and each column to a different catalytic cycle stationary point. An example is shown in Supplementary Table 1 in Supplementary Methods.

▲ **CRITICAL STEP** Take care to properly characterize the optimized geometries by examining the frequency computations (zero imaginary frequencies for minima, one imaginary frequency for transition states). The motion of this imaginary frequency (for transition states) should connect the preceding/proceeding intermediates.

▲ **CRITICAL STEP** Once the data are tabulated, the volcano plot/activity map can be automatically generated using volcanic (Box 1) instead of following the manual procedure.

Determination of linear scaling relationships ● **Timing** 1–2 h if performed manually, minutes using volcanic

- 17 Launch a plotting tool.
- 18 From the previously created spreadsheet, plot the values in each intermediate column against each other intermediate and transition state column in pairs (i.e., $\Delta G_{\text{RRS}}(\text{Int } n)$ versus $\Delta G_{\text{RRS}}(\text{Int}/\text{TS } m)$, for $n \neq m$) where **Int** *n* is the *n*th intermediate and **Int/TS** *m* the *m*th intermediate/transition state in the catalytic cycle. This process should cover all possible combinations.
- 19 Perform an ordinary least-squares linear regression of the energetic values to obtain a LFESR for the intermediate/intermediate or intermediate/transition state pair.
- 20 Establish the quality of the linear fits by calculating and comparing:
 - the average coefficient of determination (R^2)
 - the mean absolute error (MAE)
 - and the mean absolute percentage error (MAPE)For each $\Delta G_{\text{RRS}}(\text{Int } n)$, calculate the average R^2 , MAE and MAPE from all the $\Delta G_{\text{RRS}}(\text{Int } n)$ versus $\Delta G_{\text{RRS}}(\text{Int}/\text{TS } m)$ (for $n \neq m$) linear fits.

Descriptor selection ● **Timing** 1–2 h if performed manually, minutes using volcanic

- 21 Choose one (for volcano plots) or two (for activity maps) $\Delta G_{\text{RRS}}(\text{Int } n)$ quantities to serve as the descriptor variable(s). Selection should be based on the best linear fits over the complete catalytic cycle; these are deduced from the regression diagnostics determined in the preceding step. This generally corresponds to the $\Delta G_{\text{RRS}}(\text{Int } n)$ with the highest average correlation coefficient and lowest average regression error.

▲ **CRITICAL STEP** Typically, the relative free energy of one particular intermediate $\Delta G_{\text{RRS}}(\text{Int } n)$ will show the best performance. If multiple descriptors show similar diagnostics, the user should select whichever intermediate they prefer.
- 22 After selecting the descriptor variable, cast the relative stabilities of each intermediate (other than the species serving as the descriptor), **Int** *n*, and transition state, **TS** *n*, as a linear function of the descriptor. That is:

$$\Delta G_{\text{RRS}}(\text{Int}/\text{TS } n) = A \times \Delta G_{\text{RRS}}(\text{Descriptor}) + b$$

where $\Delta G_{\text{RRS}}(\text{Descriptor})$ is the chosen descriptor variable, *A* the slope and *b* the intercept.

- 23 If multiple descriptors are chosen, describe the relative stabilities of each intermediate, **Int** *n*, and transition state, **TS** *n*, as a linear function of the descriptor variables using multiple linear regressions.

For two chosen descriptor variables, the relative free energy of a intermediate/transition state will be represented by equations of the form:

$$\Delta G_{\text{RRS}}(\text{Int}/\text{TS } n) = A_1 \times \Delta G_{\text{RRS}}(\text{Descriptor } 1) + A_2 \times \Delta G_{\text{RRS}}(\text{Descriptor } 2) + b$$

where $\Delta G_{\text{RRS}}(\text{Descriptor } 1/2)$ are the descriptor variables, A_1/A_2 the slopes and *b* the intercept. The values of the coefficients and intercept need to be optimized through multiple least-squares fitting, with the resulting regression diagnosed using the same criteria of the previous step. Provided sufficient accuracy, the combination of descriptor variables can be used to build an activity map.

- 24 On the basis of the decisions made in the previous two steps (i.e., depending on the number of descriptors), the LFESRs are either postprocessed to build a volcano plot (for single descriptor) or an activity map (for two descriptors). Correspondingly, the protocol branches here: to construct a volcano plot, follow Steps 25–39; to construct an activity map, follow Steps 40–56.

▲ **CRITICAL STEP** Typically, the $\Delta G_{\text{RRS}}(\text{Int } n)$ having the best regression diagnostics with the other intermediates/transition states is chosen as the descriptor. Other parameters may also influence this decision, such as how easily the quantity can be calculated (e.g., the size of the species) or if this quantity provides a particularly good description of a critically important transition state (e.g., the rate-determining TS).

▲ **CRITICAL STEP** In principle, it is not necessary to use the relative stability of a particular intermediate as the descriptor. Other steric or electronic parameters might also be competent descriptors, but such discussions are beyond the scope of this protocol.

▲ **CRITICAL STEP** It is important to consider multiple regression diagnostics when choosing a descriptor. Considering only one parameter, such as determination coefficients (R^2), could lead to biased conclusions. For example, if the relative energy of an intermediate remains constant irrespective of the catalyst, regressing it as a function of a descriptor variable will yield a low R^2 value. However, the same descriptor will also have a low MAE.

SRP for one descriptor ● **Timing** 1–2 h if performed manually, seconds using volcanic

- 25 Estimate the relative free energies associated with moving between two (non)-linked intermediates/transition states as a function of the descriptor variable using the previously established LFESRs. Depending on the desired volcano plot variant (thermodynamic, kinetic or energy span) follow Steps 26, 27 and/or 28, respectively. The relative free energy of the separated reactants is zero by definition, while the total reaction energy is obtained from $\Delta G_{\text{rxn}} = G(\text{Products}) - G(\text{Reactants})$.

- In all cases, the ΔG_{RRS} values on the right side of the equations should be replaced with the LFESRs
- The new equations describing the free energy associated with moving between linked intermediates derived below should each be cast in terms of the descriptor variable
- By definition, the entry point of the catalytic cycle has a relative free energy of zero (i.e., $\Delta G(\text{Int } 1) = 0$). For the reaction step closing the cycle the reaction free energy (i.e., ΔG_{rxn}) must be added

- 26 *Thermodynamic volcano*. Estimate the free energy change associated with moving between intermediates directly linked in the catalytic cycle. For a cycle consisting of N intermediates (**Int**), the following equations would define the $-\Delta G$ of the individual reaction steps.

For the first step:

$$-\Delta G(\text{Int } 1 \rightarrow \text{Int } 2) = -[\Delta G_{\text{RRS}}(\text{Int } 2) - \Delta G_{\text{RRS}}(\text{Int } 1)]$$

For all intermediate steps from $n = 2$ until $n = N - 1$:

$$-\Delta G(\text{Int } n \rightarrow \text{Int } n+1) = -[\Delta G_{\text{RRS}}(\text{Int } n+1) - \Delta G_{\text{RRS}}(\text{Int } n)]$$

For the final step to close the catalytic cycle:

$$-\Delta G(\text{Int } N \rightarrow \text{Int } 1) = -[\Delta G_{\text{RRS}}(\text{Int } 1) + \Delta G_{\text{rxn}} - \Delta G_{\text{RRS}}(\text{Int } N)]$$

- 27 *Kinetic volcano*. Estimate the free energy change associated with moving between directly linked intermediates and transition states. For a cycle consisting of N intermediates (**Int**) and transition states (**TS**), the following equations would define the $-\Delta G$ of the individual reaction steps.

For the first step:

$$-\Delta G(\text{Int } 1 \rightarrow \text{TS } 1) = -[\Delta G_{\text{RRS}}(\text{TS } 1) - \Delta G_{\text{RRS}}(\text{Int } 1)]$$

For all intermediate steps from $n = 1$ until $n = N - 1$:

$$-\Delta G(\text{TS } n \rightarrow \text{Int } n+1) = -[\Delta G_{\text{RRS}}(\text{Int } n+1) - \Delta G_{\text{RRS}}(\text{TS } n)]$$

and

$$-\Delta G(\text{Int } n+1 \rightarrow \text{TS } n+1) = -[\Delta G_{\text{RRS}}(\text{TS } n+1) - \Delta G_{\text{RRS}}(\text{Int } n+1)]$$

For the final step to close the catalytic cycle:

$$-\Delta G(\text{TS } N \rightarrow \text{Int } 1) = -[\Delta G_{\text{RRS}}(\text{Int } 1) + \Delta G_{\text{rxn}} - \Delta G_{\text{RRS}}(\text{TS } N)]$$

▲ CRITICAL STEP Note that, while this kinetic volcano includes both thermodynamic and kinetic information, we strongly recommend using the use of energy span volcano (see below), which provides a more comprehensive description of catalytic activity.

- 28 **Energy span volcano.** Estimate the free energy difference associated with moving between each linked and nonlinked intermediate/transition state pair. For a cycle consisting of N intermediates (**Int**) and transition states (**TS**), the following equations would define the $-\Delta G$ of the individual reaction steps (e.g., **Int** $n \rightarrow$ **TS** m for all combinations of n and m where $1 \leq n, m \leq N$).

$$\Delta G(\text{Int } n \rightarrow \text{TS } m) = [\Delta G_{\text{RRS}}(\text{TS } m) - \Delta G_{\text{RRS}}(\text{Int } n) + \delta G_{n,m}]$$

$$\delta G_{n,m} = \begin{cases} 0, & \text{if TS } m \text{ after Int } n, \\ \Delta G_{\text{rxn}}, & \text{if TS } m \text{ before Int } n \end{cases}$$

where **TS** m and **Int** n are the m th transition state and n th intermediate in the profile, respectively, and ΔG_{rxn} is the Gibbs free energy of the reaction. Note that a correction term $\delta G_{n,m}$ is added to account for combinations in which **TS** m is before **Int** n in the catalytic cycle.

- 29 Using a plotting tool, create a single plot where each of the linear equations defining the negative of reaction energy (obtained from the proceeding steps) are plotted as a function of the descriptor variable. For a thermodynamic volcano, this would be sets of $\{-\Delta G(\text{Int } n \rightarrow \text{Int } n + 1)\}$; for a kinetic volcano, this would be sets of $\{-\Delta G(\text{Int } n \rightarrow \text{Int } n + 1)\}$, and $-\Delta G(\text{TS } n \rightarrow \text{Int } n + 1)\}$; and for an energy span volcano, this would be set of $\{-\Delta G(\text{Int } n \rightarrow \text{TS } m)\}$. Following construction, this ‘SRP’ can be used to predict the energetics associated with each step of the catalytic cycle through knowledge of the value of the descriptor variable.

▲ CRITICAL STEP By convention, the negative of the reaction energies, $-\Delta G$, are plotted versus the descriptor. Plotting the positive value of this quantity (i.e., ΔG) will lead to an inverted volcano.

Construction of the volcano plot ● Timing 1–2 h if performed manually, minutes using volcanic

- 30 Volcano plots define the most energetically costly reaction step (i.e., the potential determining step (pds) for thermodynamic, the kinetic determining step (kds) for kinetic, and the energy span (δE) for energy span volcanoes) of the catalytic cycle for a given value of descriptor variable. To obtain the volcano curve from the SRP, take the reaction line with the smallest value among all possibilities over the desired range of the descriptor variable. Mathematically, this is accomplished using the following expressions for a thermodynamic, kinetic and energy span volcanoes, respectively.

$$-\Delta G(\text{pds}) = \min.\{[-\Delta G(\text{Int } n \rightarrow \text{Int } n + 1)]\} \quad (1)$$

$$-\Delta G(\text{kds}) = \min.\{[-\Delta G(\text{Int } n \rightarrow \text{TS } n)]\} \quad (2)$$

$$-\delta E = \min.\{[-\Delta G(\text{Int } n \rightarrow \text{TS } m)]\} \quad (3)$$

In most cases, the final volcano shape will consist of multiple reaction lines as the range of descriptor values is traversed. In practice, creating the volcano plot simply involves taking the lowest line seen in the SRP for each value of the descriptor variable. Retrieval of the volcano-shaped curve marks the final step in volcano construction for the thermodynamic and kinetic volcano. If desired, the energy span volcano can be further postprocessed to generate a TOF volcano (see Steps 31–33).

(Optional) Generate a TOF volcano from an energy span volcano

- 31 To construct a TOF volcano, compute the hypothetical free energy profile from the LFESRs for values of the descriptor variable within a desired range.
- 32 Calculate the TOF corresponding to each desired descriptor variable using the hypothetical free energy profiles determined in the previous step. TOFs can be calculated using freely available software such as AUTOF⁵², or volcanic. The TOF can also be obtained using the following formula:

$$\text{TOF} = \frac{k_B T}{h} \frac{e^{-\Delta G_{\text{rxn}}/RT} - 1}{\sum_{n,m=1}^N e^{(\Delta G(\text{Int } n \rightarrow \text{TS } m))/RT}}$$

where $\Delta G(\text{Int } n \rightarrow \text{TS } m)$ is the free energy change associated with going from the n th intermediate to the m th transition state as defined in Step 28; ΔG_{rxn} is the Gibbs free energy of the reaction, k_B is the Boltzmann constant, h is the Planck constant, R is the ideal gas constant and T is the temperature.

- 33 Launch a plotting tool. Plot the descriptor variable and the corresponding TOF values (the relative activity of different catalysts are easier to see if the y -axis is plotted in log scale) to obtain the TOF volcano curve.

▲ CRITICAL STEP Often, the TOF equation can be further simplified because the sum in the denominator is frequently dominated by a single term. This leads to:

$$\text{TOF} \approx \frac{k_B T}{h} e^{-\delta E/RT}$$

where k_B is the Boltzmann constant, h is the Planck constant, R is the ideal gas constant, T is the temperature and δE is the energy span. This quantity is defined as

$$\delta E = \max. [\{\Delta G_{\text{RRS}}(\text{TS } m) - \Delta G_{\text{RRS}}(\text{Int } n) + \delta G_{n,m}\}]$$

$$\delta G_{n,m} = \begin{cases} 0, & \text{if TS } m \text{ after Int } n, \\ \Delta G_{\text{rxn}}, & \text{if TS } m \text{ before Int } n \end{cases}$$

where **TS** m and **Int** n are the m th TS and n th intermediate in the profile, respectively, and ΔG_{rxn} is the Gibbs free energy of the reaction. This approximate TOF formulation implies that the turnover depends only on the largest energy difference in the catalytic cycle (i.e., the energy span, δE , between the most populated intermediate and the rate-determining transition state). Thus, a TOF volcano can be constructed directly from the energy span volcano. The user should verify that the full and the approximate TOF values agree with one another. Any disagreement implies that more than one transition state and/or intermediate (ignored in the approximated TOF calculation) contributes substantially to the TOF.

▲ CRITICAL STEP Computing the hypothetical free energy profiles of descriptor values in small increments (<1 kcal/mol) will lead to smoother TOF volcano. This is particularly true in areas where there is a change in curvature. In some cases, smaller step sizes may be required in areas where the curvature of the plot changes.

▲ CRITICAL STEP The TOF value is plotted in the logarithmic scale in the y -axis to resemble the corresponding energy span volcano. Plotting the y -axis at a linear scale shows the TOF curve (generally) as a very sharp peak.

Analysis and interpretation of volcano plots ● Timing 1–24 h, depending on the complexity of the volcano

- 34 Add points corresponding to catalysts to the volcano plot using the value of the descriptor variable as the x -coordinate and the $-\Delta G(\text{pds/kds})$ or $-\delta E$ as the y -coordinate.
- 35 Inspect the relative positioning of the catalysts. The anticipated most active candidates are found highest on the plot, typically on or near the volcano summit or plateau. Less active catalysts lie lower on the plot, often along the left and right slopes.
- 36 Splitting the volcano plot into sections (by drawing vertical lines) based on the most difficult reaction step in the catalytic cycle can be helpful for understanding the factors that limit catalytic activity. In the volcano curve, a change of slope corresponds to a change in the nature of the most difficult reaction step (for thermodynamic and kinetic volcanoes) or TDI/TDTS pair (for energy span volcanoes).
- 37 By definition, the TOF volcano is a continuous curve. Information about the TDI/TDTS pair that control the TOF can be obtained from splitting the related energy span volcano into sections. To find the position of the vertical lines that partition the TOF volcano into regions, split the corresponding energy span volcano and transfer the position of the boundaries to the TOF volcano.
- 38 Rationalize the observed trends in catalytic behaviour by comparing the chemical compositions of the catalysts and/or by comparing the identity of the reactive centers/surroundings.
- 39 If the volcano is being used for screening prospective catalysts, compute the descriptor variable for each new candidate. The location of each point on the volcano plot then corresponds to the value of the descriptor variable (x -coordinate) and $-\Delta G(\text{pds/kds})$ or $-\delta E$ (y -coordinate). The value of the y -coordinate is obtained from the equation of the volcano curve line associated with the

region of the volcano where the catalyst is located, which is determined from the value of the descriptor variable.

SRP for two descriptors ● **Timing** 1–2 h if performed manually, minutes using volcanic

▲ **CRITICAL** This step should be performed after Step 24 if two descriptor variables are chosen.

- 40 Estimate the relative free energies associated with moving between two (non)-linked intermediates/transition states as a function of the two descriptor variables using the previously established multiple LFESRs (see Step 23). Depending on the desired activity map variant (thermodynamic, kinetic or energy span), follow Step 41, 42 or 43, respectively. The relative free energy of the separated reactants is zero by definition, while the total reaction energy is obtained from $\Delta G_{\text{rxn}} = G(\text{Products}) - G(\text{Reactants})$.
 - In all cases, the ΔG_{RRS} values on the right side of the equations should be replaced with the LFESRs
 - The new equations describing the free energy associated with moving between linked intermediates derived below should each be cast in terms of the two descriptor variables
 - Note that, by definition, the entry point of the catalytic cycle has a relative free energy of zero (i.e., $\Delta G(\text{Int } 1) = 0$). For the reaction step closing the cycle the reaction free energy (i.e., ΔG_{rxn}) must be added
- 41 *Thermodynamic activity map*. Estimate the free energy change associated with moving between intermediates directly linked in the catalytic cycle. For a cycle consisting of N intermediates (Int), the following equations would define the $-\Delta G$ of the individual reaction steps.

For the first step:

$$-\Delta G(\text{Int } 1 \rightarrow \text{Int } 2) = -[\Delta G_{\text{RRS}}(\text{Int } 2) - \Delta G_{\text{RRS}}(\text{Int } 1)]$$

For all intermediate steps from $n > 1$ until $< N$:

$$-\Delta G(\text{Int } n \rightarrow \text{Int } n + 1) = -[\Delta G_{\text{RRS}}(\text{Int } n + 1) - \Delta G_{\text{RRS}}(\text{Int } n)]$$

For the final step to close the catalytic cycle:

$$-\Delta G(\text{Int } N \rightarrow \text{Int } 1) = -[\Delta G_{\text{RRS}}(\text{Int } 1) + \Delta G_{\text{rxn}} - \Delta G_{\text{RRS}}(\text{Int } N)]$$

- 42 *Kinetic activity map*. Estimate the free energy change associated with moving between directly linked intermediates and transition states. For a cycle consisting of N intermediates (Int) and transition states (TS), the following equations would define the $-\Delta G$ of the individual reaction steps.

For the first step:

$$-\Delta G(\text{Int } 1 \rightarrow \text{TS } 1) = -[\Delta G_{\text{RRS}}(\text{TS } 1) - \Delta G_{\text{RRS}}(\text{Int } 1)]$$

For all intermediate steps from $n = 2$ until $n = N - 1$:

$$-\Delta G(\text{TS } n \rightarrow \text{Int } n + 1) = -[\Delta G_{\text{RRS}}(\text{Int } n + 1) - \Delta G_{\text{RRS}}(\text{TS } n)]$$

and

$$-\Delta G(\text{Int } n + 1 \rightarrow \text{TS } n + 1) = -[\Delta G_{\text{RRS}}(\text{TS } n + 1) - \Delta G_{\text{RRS}}(\text{Int } n + 1)]$$

For the final step to close the catalytic cycle:

$$-\Delta G(\text{TS } N \rightarrow \text{Int } 1) = -[\Delta G_{\text{RRS}}(\text{TS } N) - \Delta G_{\text{RRS}}(\text{Int } 1) - \Delta G_{\text{rxn}}]$$

▲ **CRITICAL STEP** Note that, while this kinetic activity map includes both thermodynamic and kinetic information, we strongly recommend using the use of energy span activity map (see below), which provides a more comprehensive description of catalytic activity.

- 43 *Energy span activity map*. Estimate the free energy difference associated with moving between each linked and nonlinked intermediate/transition state pair.

For a cycle consisting of N intermediates (**Int**) and transition states (**TS**), the following equations would define the $-\Delta G$ of the individual reaction steps (e.g., **Int** $n \rightarrow$ **TS** m for all combinations of n and m where $1 \leq n, m \leq N$).

$$\Delta G(\text{Int } n \rightarrow \text{TS } m) = [\Delta G_{\text{RRS}}(\text{TS } m) - \Delta G_{\text{RRS}}(\text{Int } n) + \delta G_{n,m}]$$

$$\delta G_{n,m} = \begin{cases} 0, & \text{if TS } m \text{ after Int } n, \\ \Delta G_{\text{rxn}}, & \text{if TS } m \text{ before Int } n \end{cases}$$

where **TS** m and **Int** n are the m th TS and n th intermediate in the profile, respectively, and ΔG_{rxn} is the Gibbs free energy of the reaction. Note that a correction term $\delta G_{n,m}$ is added to account for combinations in which **TS** m is before **Int** n in the catalytic cycle.

- 44 Using a plotting tool, draw a single plot overlaying all the surfaces corresponding to the negative of all reaction energies (obtained in Steps 41, 42 and 43 for thermodynamic, kinetic and energy span activity maps, respectively) described as a function of the two descriptor variables. These surfaces would be the set of $\{-\Delta G(\text{Int } n \rightarrow \text{Int } n + 1)\}$ (for thermodynamic activity maps) or $\{-\Delta G(\text{Int } n \rightarrow \text{TS } n) / -\Delta G(\text{TS } n \rightarrow \text{Int } n + 1)\}$ (for kinetic activity maps). In energy span activity maps, these surfaces are the set of $\{-\Delta G(\text{Int } n \rightarrow \text{TS } m)\}$.

Construction of the activity map ● Timing 2–4 h if performed manually, minutes using volcanic

- 45 Define a 2D grid spanning the desired range of the two chosen descriptors. The range should encompass the maximum span of the two descriptor variables in the initial pool.
- 46 Determine the most energetically costly step ($-\Delta G(\text{pds}) / -\Delta G(\text{kds}) / -\delta E$) at each point on the grid, which can be obtained from the following expression for a thermodynamic, kinetic and energy span activity map, respectively.

$$-\Delta G(\text{pds}) = \min.\{[-\Delta G(\text{Int } n \rightarrow \text{Int } n + 1)]\} \quad (4)$$

$$-\Delta G(\text{kds}) = \min.\{[-\Delta G(\text{Int } n \rightarrow \text{TS } n)]\} \quad (5)$$

$$-\delta E = \min.\{[-\Delta G(\text{Int } n \rightarrow \text{TS } m)]\} \quad (6)$$

The reaction surface with the most negative (or least positive) values defines the pds/kds/energy span for that region of the two descriptor variables.

- 47 Plot a 2D heatmap with the two chosen descriptors as the x - and y - axes, and the $-\Delta G(\text{pds}) / -\Delta G(\text{kds}) / -\delta E$ as a color-coded z -axis. The energy span activity map can be further postprocessed to generate a TOF activity map (Steps 48–50), if desired.
- 48 To generate a TOF activity map, compute the hypothetical free energy profile (from the previously established LFESRs) for each grid point (i.e., any values in the range of the two descriptor variables).
- 49 Calculate the TOF corresponding to this hypothetical free energy profile within the framework of energy span model for each grid point. TOFs can be calculated using freely available software such as AUTOF⁵², or volcanic. The TOF can also be obtained using the following formula,

$$\text{TOF} = \frac{k_{\text{B}} T}{h} \frac{e^{-\Delta G_{\text{rxn}}/RT} - 1}{\sum_{n,m=1}^N e^{(\Delta G(\text{Int } n \rightarrow \text{TS } m))/RT}}$$

where $\Delta G(\text{Int } n \rightarrow \text{TS } m)$ is the free energy change associated with going from the n th intermediate to the m th transition state as defined in Step 43; ΔG_{rxn} is the Gibbs free energy of the reaction, k_{B} is the Boltzmann constant, h is the Planck constant, R is the ideal gas constant and T is the temperature.

- 50 Launch a plotting tool. Plot the two descriptor variables (x - and y -axis) and the corresponding TOF values (z -axis) to obtain the TOF activity map. Note that the relative activity of different catalysts are easier to see when the y -axis is plotted in log scale.

Analysis and interpretation of activity maps ● Timing 1–24 h, depending on the complexity of the activity map

- 51 Separate the activity map into different regions associated with different reaction steps or TDI/TDTS pairs the correspond to the most energetically costly parts of the catalytic cycle. Determine the identity of this most energetically costly step at each grid point.

- 52 Color each grid point according to the identity of the most energetically costly step, and locate the boundaries between the different regions.
- 53 Locate the most active region in the activity map (associated with the lowest value of $-\Delta G(\text{pds})/ -\Delta G(\text{kds})/ -\delta E$ and identify the corresponding reaction step.
- 54 Add points for each catalyst to the activity map using the descriptor variables as the x - and y -coordinates.
- 55 Inspect the relative positioning of the catalysts. Active species will appear near the most active region of the activity map.
- 56 Rationalize the observed trends in catalytic behaviour by comparing catalyst composition (reactive centers/ligands).

Troubleshooting

Step 16: Problem: missing data

Cause: Specific transition states may be difficult to converge.

Solution: If a single transition state energy is missing for a given reaction profile, it should be possible to predict its value from the corresponding LFESRs in a later stage (see Steps 21–23).

Step 24: Problem: inaccurate scaling relationship

Cause: If no acceptable linear fits can be found, unintended factors (e.g., new types of intramolecular interactions) may be causing disparity among the catalyst pool that cause description with a single variable to be insufficient.

Solution: In such cases, construct an activity map that includes an additional descriptor.

Step 24: Problem: little or no improvement seen when comparing results using two descriptor variables with the single-variable volcano plot

Cause: Co-linearity of the selected pair may mean that there is little to no additional information in the activity plot compared with the volcano plot.

Solution: An exhaustive test of all the possible combinations, as described in Steps 18–20 for simple linear regression, will lead to identification of the best possible combination.

Timing

Steps 1–2, identification of the reaction mechanism: if discovery is required, days to weeks, depending on the number of steps in the catalytic cycle and on the existence of previous literature reports

Steps 3–4, selection of the pool of catalysts: minutes to hours

Steps 5–16, reaction data acquisition: one to several weeks, depending on catalyst size and the number of catalytic cycle intermediates and transition states (this is the most time-consuming step in creating volcano plots/activity maps; the timing assumes sufficient computational facilities and a cost of 10^4 – 10^5 CPU h)

Steps 17–20, determination of LFESRs: 1–2 h if performed manually, minutes using volcanic

Steps 21–24, descriptor selection: 1–2 h if performed manually, minutes using volcanic

Steps 25–29, SRP for one descriptor: 1–2 h if performed manually, minutes using volcanic

Steps 30–33, construction of the volcano plot: 1–2 h if performed manually, minutes using volcanic

Steps 34–39, analysis and interpretation of volcano plots: 1–24 h, depending on the complexity of the volcano

Steps 40–44, SRP for two descriptors: 1–2 h if performed manually, minutes using volcanic

Steps 45–50, construction of the activity map: 2–4 h if performed manually, minutes using volcanic

Steps 51–56, analysis and interpretation of activity maps: 1–24 h, depending on the complexity of the activity map

Anticipated results

Having described the protocol for constructing volcano plots and activity maps, we now demonstrate their utility by analyzing two representative chemical processes: (1) the hydrogenation of CO_2 to formate catalyzed by metal pincer complexes (volcano plots) and (2) the energetics of C–C cross-coupling reactions (activity maps). Below, we demonstrate the complete protocol, detail expected results and reenact the entire process using our automated postprocessing tool (volcanic) from the

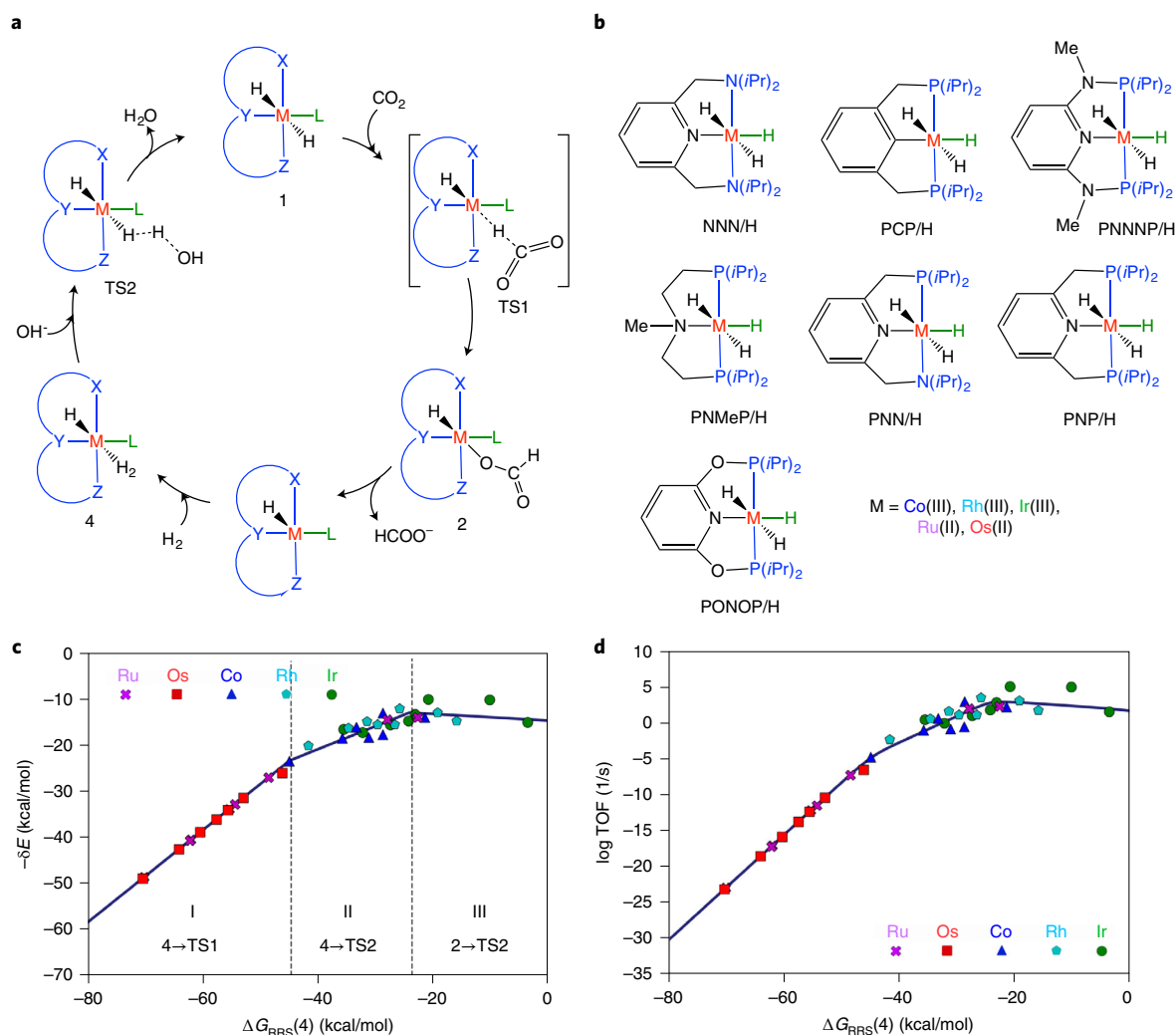


Fig. 4 | Volcano plot study of CO₂ hydrogenation with transition metal pincer complexes. **a**, Catalytic cycle for the transition metal pincer complex-catalyzed hydrogenation of CO₂ to formate. **b**, Chemical composition of the catalyst pool used to construct the LFESRs and the volcano plots. **c,d**, Energy span (**c**) and TOF volcano plots (**d**) illustrating the anticipated catalytic activity for the reaction.

raw data obtained from DFT computations. The complete worked-out example for generating the results in Fig. 4 is available in Supplementary Methods.

CO₂ hydrogenation cast into volcano plots

Transition metal pincer complexes have shown remarkable catalytic activity for hydrogenation of CO₂ to formate. To better understand the factors that influence catalytic activity, we created a series of volcano plots^{26,34}. Using the catalytic cycle depicted in Fig. 4a, we tested catalysts containing five different metals (Co, Rh, Ir, Ru and Os) combined with seven pincer ligands (Fig. 4b), for a total of 35 prospective candidates. The diversity of the catalyst pool is ensured by varying the identity of the metal center (exploring multiple periods and groups), the connecting atoms (N and P), the rigidity of the pincer backbone and their linker atoms (C, N and O).

The relative free energies of all catalytic cycle stationary points were estimated with respect to species **1** (Fig. 4a). Based on the regression diagnostics of each pair of ΔG_{RRS} combinations, $\Delta G_{RRS}(4)$ was chosen as the descriptor variable (for details on regression diagnostics, see Supplementary Information). Chemically, this descriptor corresponds to the differential binding ability between an anionic species (hydride) and a neutral species (H₂) for a given catalyst.

Utilizing the LFESRs, the catalytic cycle can be cast into an SRP by plotting the free energy change between any two intermediates and transition states as a linear function of the descriptor, as detailed

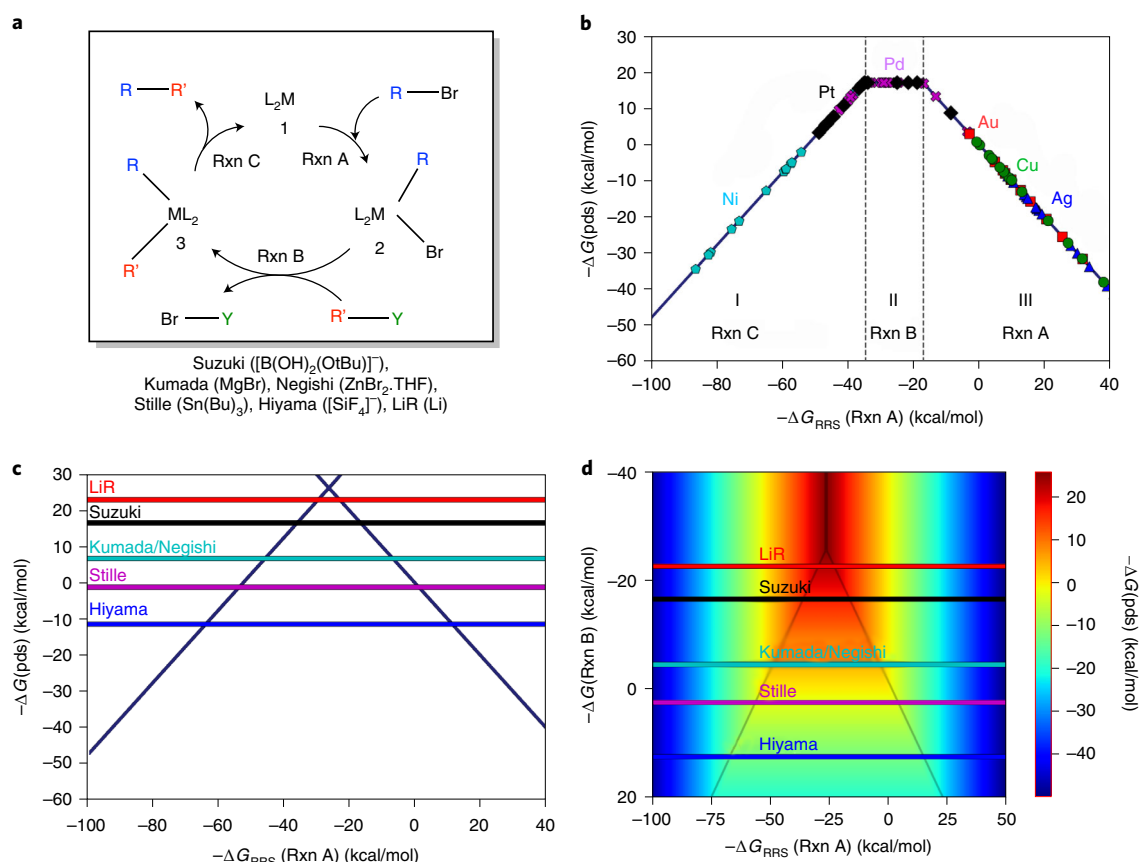


Fig. 5 | Volcano plot and activity map study of C–C cross-coupling reactions. **a**, Simplified catalytic cycle of a transition metal catalyzed cross-coupling reaction. **b**, Thermodynamic volcano plot for the Suzuki–Miyaura reaction with points colored by the identity of the metal centers. **c**, Volcano plot depicting changes in the energy needed to complete the transmetalation step (horizontal lines) for the six cross-coupling protocols. **d**, Activity map describing the entire C–C cross coupling range. **d**, Activity map describing the entire C–C cross coupling range. The lines are perfectly horizontal because the reaction energies that define the y-axis are constant irrespective of the choice of catalyst.

in Step 28. The volcano can then be constructed by taking the lowest-lying lines or, mathematically, by using Eq. 3. For the interested reader, detailed derivations of the LFESRs, SRPs and the different volcano plots are available in Supplementary Methods.

A comprehensive picture of catalytic cycle energetics is provided by the energy span volcano (Fig. 4c), which examines all intermediate/transition state pairs to determine the energy span of the catalytic cycle. The measure of catalytic activity that appears on the volcano y-axis is δE (as defined in Eq. 3), where less negative values indicate higher activity. The resulting volcano has three different TDI/TDTS combinations (2/TS2, 4/TS2 and 4/TS1) that define the energy span of the catalytic cycle. The LFESRs can also be directly postprocessed (Steps 31–33) to obtain the corresponding TOF volcano (Fig. 4d), which provides a measure of activity more intuitive to experimentalists. Points for each of the 35 catalysts can then be added to the volcano plots. Analyzing these point reveals that group 9 metals combined with π -acidic ligands have the best profiles, with activity increasing when switching from cobalt to rhodium to iridium. Note that similar conclusions regarding the underlying chemistry can be reached from the two volcano plot variants (provided in Supplementary Information), albeit the energy span and TOF volcanoes lead to a more accurate picture.

C–C cross-coupling cast into volcano plots and activity maps

Transition metal-catalyzed C–C cross-coupling reactions are commonly employed synthetic techniques for creating new C–C bonds. Several variants of this reaction class exist (e.g., Suzuki, Kumada and Negishi), which differ in the transmetalation agent used to transfer ‘R’ groups onto the catalyst (Y, Fig. 5a). Below we outline a step-by-step protocol based on volcano plots and activity maps that describes the different cross-coupling variants. The basic mechanistic steps, oxidative addition (Rxn

A), transmetalation (Rxn B) and reductive elimination (Rxn C) are conserved across all cross-coupling catalytic cycles, while the chemical agent used during the transmetalation step (Y) differs.

The Fig. 5b volcano describes the energetics of the Suzuki–Miyaura cross-coupling of a vinyl boronic acid and vinylbromide to form butadiene. The volcano was constructed from the catalytic cycle depicted in Fig. 5a using LFESRs obtained by examining 36 catalysts comprising six metals (Ni, Pd, Pt, Cu, Ag and Au) with six model ligands (CO, NH₃, PMe₃, acetone, an *N*-heterocyclic carbene and PMe₃/NH₃) using $\Delta G_{RRS}(2)$ as the descriptor variable.

Separating the volcano into different regions reveals that catalysts falling in the leftmost region are limited by energy of reductive elimination, while those falling in the rightmost region are limited by the energy of oxidative addition (Rxn A). Species located in the central region are the most thermodynamically balanced, being limited by the energy of transmetalation (Rxn B). Here, many experimental observed trends are reproduced, with palladium species located on the volcano plateau, while nickel and platinum catalysts are shifted along the left slope and the coinage metal catalysts (Cu, Ag and Au) are shifted far from the plateau along the right slope.

As mentioned above, numerous C–C cross-coupling variants (Suzuki–Miyaura, Negishi, Hiyama, Kumada and Stille) exist, each of which employs a different transmetalation agent. Reexamining Fig. 5b reveals that the central plateau region is dictated by the transmetalation energy (Rxn B). Thus, each of the cross-coupling variants is related by the energy of the transmetalation step (Fig. 5c). By including this quantity as a second descriptor variable (Rxn B), we created an activity map (Fig. 5d) that provides a holistic picture of C–C cross-coupling reactions. This map can be used both to rationalize experimentally obtained trends and to devise design strategies for optimizing catalytic activity. For example, placing points for the individual catalysts onto the activity map reveals that palladium catalysts are optimally tuned for all cross-coupling variants (not just Suzuki–Miyaura, as would be ascertained by examining only the two-dimensional volcano plot). Switching to less-reactive variants (moving toward more positive Rxn B values, *y*-axis) lowers the volcano plateau, which causes a larger array of catalysts to have ideal thermodynamic profiles. This means a plethora of nickel and even some coinage metal catalysts become thermodynamically ideal when moving to Hiyama coupling. However, there is a trade-off, as lowering the plateau also causes a decrease in thermodynamic drive of the catalytic process.

Data availability

All data to reproduce all the figures of this work are available at <https://github.com/lcmd-epfl/volcanic>, as well as instructions to generate such plots using the volcanic package.

Code availability

The volcanic package is available at <https://github.com/lcmd-epfl/volcanic>. Supplementary Methods detail the manual procedure for the application highlighted in Fig. 4.

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All authors contributed to the conceptualization, elaboration of the content and writing of the manuscript.

Competing interests

The authors declare no competing interests.

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